

ERMELO, South Africa

WMO: 682670

Lat: 26.4978S

Lon: 29.9839E

Elev: 5820

StdP: 11.86

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.5	35.3	7.4	10.0	57.3	11.3	12.1	54.6	26.9	60.8	23.5	58.4	8.0	270	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.1	83.3	58.3	81.2	58.4	79.2	58.7	65.5	74.8	64.6	73.8	63.7	72.5	11.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	106.5	68.1	61.9	103.0	67.3	61.0	99.8	66.5	33.9	74.8	33.1	74.1	32.3	72.6	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	20.5	18.2	27.9	87.3	2.3	3.0	26.3	89.5	24.9	91.2	23.6	92.9	22.0	95.2		
(5)			24.7	67.3	2.8	0.7	22.6	67.9	21.0	68.3	19.4	68.7	17.4	69.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	65.5	65.2	63.6	59.5	55.0	51.0	50.1	54.9	60.0	61.9	63.7	65.2	(6)
(7)		DBStd	7.35	4.01	3.49	3.96	4.52	4.61	4.47	4.82	6.13	6.95	6.48	5.48	4.18	(7)
(8)		HDD50	159	0	0	0	3	14	38	59	26	12	5	2	0	(8)
(9)		HDD65	2256	41	35	70	171	310	419	462	313	173	131	84	47	(9)
(10)		CDD50	3667	482	427	422	287	169	69	62	179	312	375	412	471	(10)
(11)		CDD65	286	57	42	26	5	0	0	0	1	22	36	44	53	(11)
(12)		CDH74	2485	336	252	190	50	10	0	1	41	367	476	381	381	(12)
(13)		CDH80	301	30	19	12	2	0	0	0	0	41	92	55	50	(13)
(14)	Wind	WSAvg	9.7	9.3	8.8	8.6	8.4	8.7	9.2	9.4	10.5	11.6	11.2	10.9	9.8	(14)
(15)	Precipitation	PrecAvg	27.5	4.9	3.0	3.2	1.3	0.4	0.2	0.2	0.4	0.7	3.4	4.1	5.4	(15)
(16)		PrecMax	37.7	8.3	9.4	7.3	4.3	2.1	1.5	1.7	2.0	3.2	8.5	8.0	11.9	(16)
(17)		PrecMin	19.8	1.5	0.7	0.5	0.0	0.0	0.0	0.0	0.0	0.0	1.3	1.3	2.6	(17)
(18)		PrecStd	4.8	1.7	1.8	1.7	1.0	0.5	0.4	0.4	0.6	0.7	1.7	1.7	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.6	82.5	81.8	78.8	75.3	70.7	70.9	77.6	83.9	86.3	85.3	84.5	(19)
(20)			MCWB	61.4	60.6	59.5	57.5	52.8	47.6	46.0	51.3	53.4	56.4	58.9	61.1	(20)
(21)		2%	DB	80.3	79.7	78.7	75.5	71.1	67.3	67.3	74.9	81.0	82.7	81.1	81.2	(21)
(22)			MCWB	61.8	61.8	59.2	56.1	50.9	46.7	45.8	50.5	52.5	56.1	59.4	61.5	(22)
(23)		5%	DB	77.6	77.2	75.9	73.1	68.4	64.6	64.8	72.0	78.6	79.4	78.5	78.5	(23)
(24)			MCWB	62.4	61.5	59.2	55.6	49.8	45.7	45.0	49.3	52.3	56.2	60.1	61.8	(24)
(25)		10%	DB	75.2	74.7	73.5	70.1	66.1	62.1	62.2	68.6	75.6	76.2	75.3	75.6	(25)
(26)			MCWB	62.4	61.5	59.5	55.3	49.2	45.1	44.3	47.9	51.5	56.0	59.8	61.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.8	66.0	65.5	62.2	56.6	51.3	50.4	55.7	59.3	62.8	65.0	66.6	(27)
(28)			MCDWB	75.7	75.5	74.1	70.3	65.8	62.9	62.5	69.4	73.1	74.5	75.2	77.5	(28)
(29)		2%	WB	65.4	64.8	63.8	60.2	54.4	49.6	48.4	53.7	56.6	60.7	63.5	65.0	(29)
(30)			MCDWB	74.0	74.0	71.7	68.3	64.0	60.3	60.8	67.7	71.0	73.2	74.1	75.1	(30)
(31)		5%	WB	64.4	63.8	62.5	58.8	52.8	48.3	47.0	51.8	55.0	59.3	62.4	63.9	(31)
(32)			MCDWB	73.0	72.5	69.8	66.1	62.5	59.1	59.2	65.0	70.1	71.3	72.8	73.5	(32)
(33)		10%	WB	63.4	62.9	61.3	57.4	51.4	47.0	45.7	50.2	53.6	57.9	61.2	62.8	(33)
(34)			MCDWB	71.6	70.6	68.2	65.5	61.2	56.9	57.6	62.3	68.6	69.6	71.2	72.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.1	17.2	17.9	19.2	21.6	22.1	22.9	24.2	25.0	22.7	19.8	19.0	(35)
(36)			MCDBR	22.0	22.0	22.7	23.1	24.0	25.0	25.8	27.1	28.3	28.1	24.5	23.5	(36)
(37)			MCWBR	8.1	8.2	8.0	8.5	9.2	10.5	10.9	10.0	9.6	9.2	8.7	8.8	(37)
(38)		5% WB	MCDBR	18.3	17.8	17.8	18.9	20.6	21.2	22.5	23.8	25.3	23.0	21.0	19.8	(38)
(39)			MCWBR	7.8	7.7	7.8	8.0	9.1	9.9	10.1	9.3	10.4	9.7	8.8	9.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.383	0.373	0.367	0.326	0.288	0.271	0.271	0.328	0.374	0.418	0.381	0.388	(40)	
(41)		taud	2.314	2.343	2.346	2.428	2.512	2.530	2.514	2.348	2.249	2.168	2.303	2.293	(41)	
(42)		Ebn at Noon	304	302	294	292	292	291	296	287	287	285	303	304	(42)	
(43)		Edh at Noon	44	42	39	33	28	26	27	35	43	49	44	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2005	1955	1704	1457	1335	1220	1314	1529	1764	1836	1963	2038	(44)	
(45)		RadStd	193	178	156	115	71	65	42	81	103	130	169	114	45	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	-326	+421	+157

Nomenclature: See separate page